Lecture 7: Confidence Intervals

Sampling Distributions · Standard Errors · Interval Estimation

Previously, on Lectures 5 & 6...

MLE: $\hat{\theta} = \arg\max \ell(\theta)$. The go-to method for point estimation.

Asymptotic normality: $\sqrt{n}(\hat{\theta} - \theta_0) \to N(0, 1/I(\theta_0))$. MLE is approximately Normal.

Fisher info: $I(\theta) = -E[\ell''(\theta)]$. Measures how much one obs tells us.

CR bound: $\mathrm{Var}(\hat{\theta}) \geq 1/(nI(\theta))$. A floor on uncertainty.

MAP: $\hat{\theta} = \arg\max[\ell(\theta) + \log P(\theta)]$. MLE + prior = regularization.

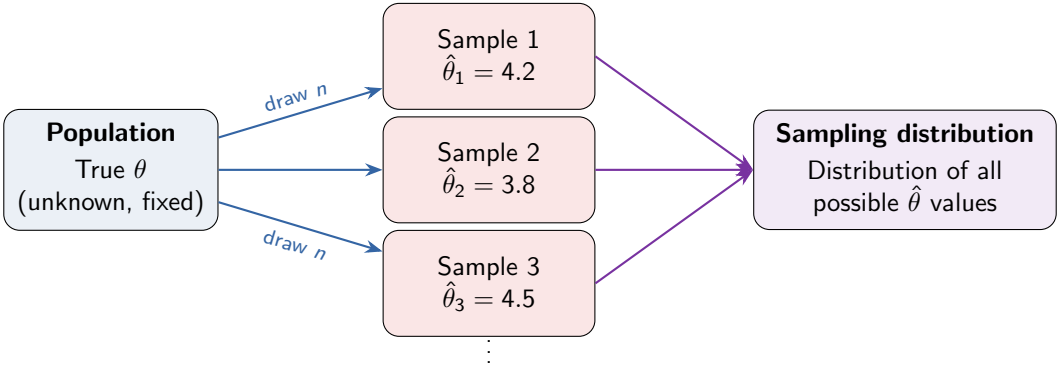
Today: A point estimate is just a number. How **uncertain** is it?

"52% support candidate A" means nothing without $\pm$ something.

Sampling Distributions

Your estimator is a random variable

The Key Idea: $\hat{\theta}$ Is Random



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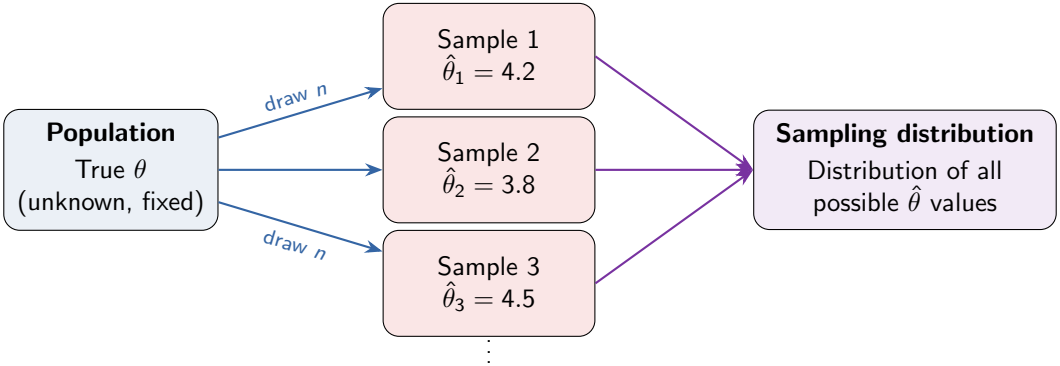

Each sample gives a **different** estimate. The **sampling distribution** describes this variability.

The Sampling Distribution of $\bar{X}$

If $X_1, \ldots, X_n \overset{\text{iid}}{\sim} (\mu, \sigma^2)$, what is the distribution of $\bar{X} = \frac{1}{n} \sum X_i$?

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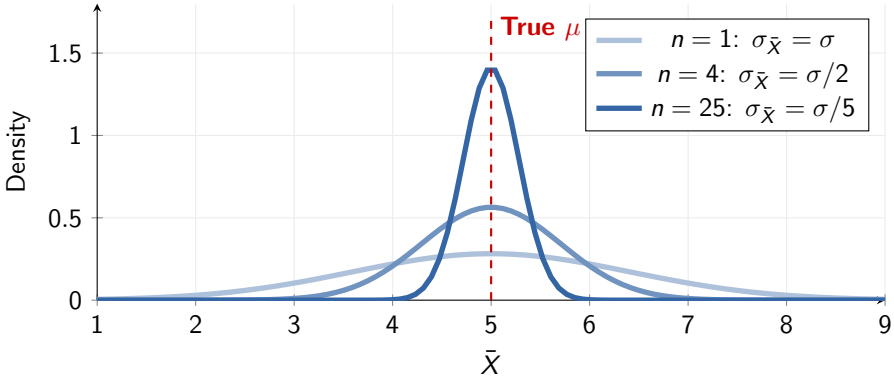

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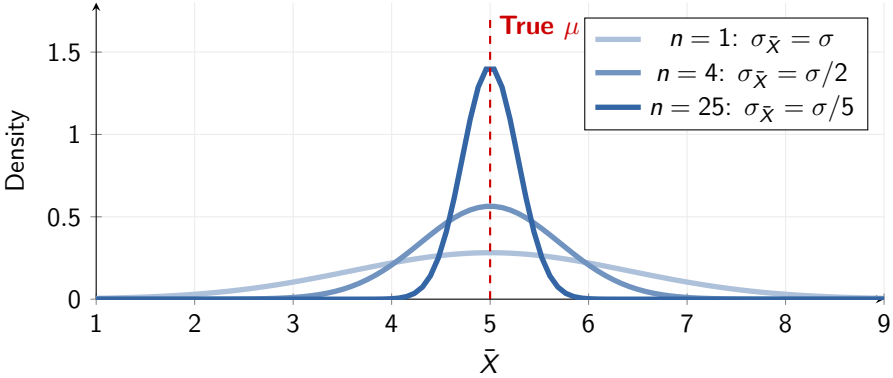

$\bar{X} \sim N\left(\mu,\ \frac{\sigma^2}{n}\right)$ (exactly if Normal; approximately by CLT otherwise)

The $\sqrt{n}$ Law

$$\mathrm{SE}(\bar{X}) \quad = \quad \frac{\sigma}{\sqrt{n}}$$

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To **halve** the standard error, you must **quadruple** the sample size.
Precision is expensive! Diminishing returns from more data.

Standard Error: Known vs Estimated

If σ is known:

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Rare in practice.
Use the z-distribution.

General pattern for any MLE (from Lecture 5):

$\widehat{\mathrm{SE}}(\hat{\theta}_{\mathrm{MLE}}) = \frac{1}{\sqrt{n \cdot I(\hat{\theta})}}$ — plug in $\hat{\theta}$ for θ in the Fisher information.

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Confidence Intervals

From point estimate to interval estimate

What Is a Confidence Interval?

A **95% confidence interval** is a random interval $[L, U]$ such that

$$P(L \leq \theta \leq U) = 0.95$$

Before you collect data, there is a 95% chance the interval will contain θ.

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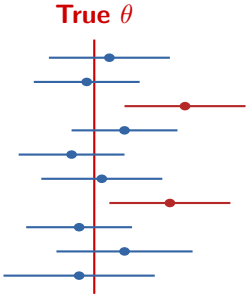

10 simulated 95% CIs

2 miss θ (expected $\approx$ 1 in 20)

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For 95%: $\hat{\theta} \pm 1.96 \cdot \mathrm{SE}$ For 99%: $\hat{\theta} \pm 2.576 \cdot \mathrm{SE}$ For 90%: $\hat{\theta} \pm 1.645 \cdot \mathrm{SE}$

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Recipe:

1. Compute the point estimate $\hat{\theta}$
2. Compute (or estimate) the standard error $\mathrm{SE}(\hat{\theta})$
3. Choose the confidence level $\Rightarrow$ look up $z_{\alpha/2}$
4. Report: $\hat{\theta} \pm z_{\alpha/2} \cdot \mathrm{SE}$

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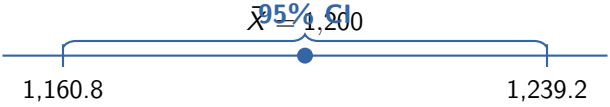
Step 4:

$$\bar{X} \pm 1.96 \cdot \text{SE} = 1{,}200 \pm 1.96 \times 20 = 1{,}200 \pm 39.2$$



"We are 95% confident that the true mean lifetime is between 1,161 and 1,239 hours."

What 95% Confidence **Really** Means

Correct:

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The *procedure* works 95% of the time.

Analogy: A 95%-accurate archer hits the target 95% of the time. After the arrow lands, it either hit or missed — the probability is gone.

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What Determines the Width?

$$\text{Width} = 2 \cdot z_{\alpha/2} \cdot \frac{S}{\sqrt{n}}$$

Confidence level $\uparrow$

$z_{\alpha/2}$ increases

$\Rightarrow$ **wider** CI

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Sample size n $\uparrow$

$\sqrt{n}$ in denominator

$\Rightarrow$ **narrower** CI

But diminishing returns ($\sqrt{n}$ law!)

CIs for Proportions

Polls, A/B tests, and defect rates

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Example: Poll of $n = 1{,}000$ voters, $\hat{p} = 0.52$.

$$\text{SE} = \sqrt{\frac{0.52 \times 0.48}{1000}} = 0.0158 \qquad \text{CI: } 0.52 \pm 0.031 = [0.489,\ 0.551]$$

The race is too close to call! The CI includes 0.5.

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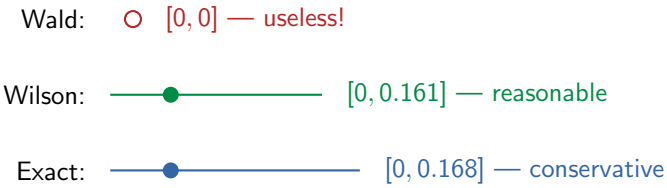

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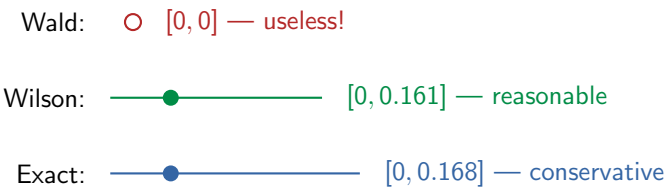

Rule of thumb: Wald works well when $n\hat{p} \geq 5$ and $n(1-\hat{p}) \geq 5$.
For small n or $\hat{p}$ near 0 or 1, use the **Wilson** interval instead.

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Wald: centered at $\hat{p}$ $\xrightarrow{\text{shrink toward } \frac{1}{2}}$ Wilson: centered at $\tilde{p}$

Wilson is the default in many statistics packages. Wald is mostly for teaching.

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Solve for n (use worst case $p = 0.5$):

$$n \geq \left(\frac{z_{\alpha/2}}{\text{ME}}\right)^2 \cdot p(1-p) = \left(\frac{1.96}{0.03}\right)^2 \cdot 0.25 = 1{,}068$$

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$\text{ME} = \pm 5\%$ $n = 385$	$\text{ME} = \pm 3\%$ $n = 1{,}068$	$\text{ME} = \pm 1\%$ $n = 9{,}604$

The $\sqrt{n}$ law again: $10\times$ more precision requires $100\times$ more data.

Beyond Wald

t-intervals, likelihood intervals, and the Bayesian alternative

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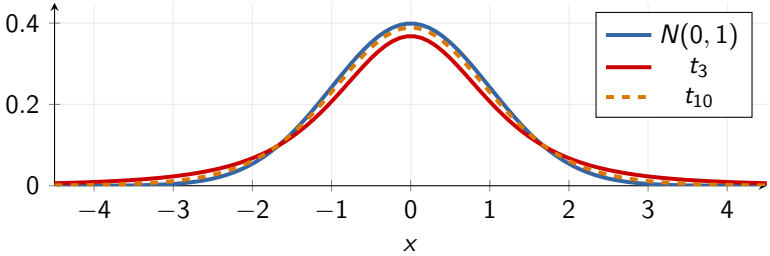

As $n \to \infty$: $t_{n-1} \to N(0,1)$. For $n \geq 30$, the difference is negligible.

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Concrete: $n = 50$ earthquakes/year, $\hat{\lambda} = 3.2$. SE $= \sqrt{3.2/50} = 0.253$. CI: $[2.70,\ 3.70]$.

Confidence Intervals vs Credible Intervals

Confidence Interval (Frequentist)

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No prior needed.

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With large n, they often give nearly identical intervals.
The philosophical difference matters most with small n or strong priors.

Neither is "wrong" — they answer **different questions**.

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$$g(\hat{\theta}) \quad \dot{\sim} \quad N\big(g(\theta), \quad [g'(\theta)]^2 \cdot \sigma^2_{\hat{\theta}}\big)$$

$$\text{SE of } g(\hat{\theta}) \quad \approx \quad |g'(\hat{\theta})| \cdot \text{ SE of } \hat{\theta}$$

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$$\text{SE of } g(\hat{\theta}) \quad \approx \quad |g'(\hat{\theta})| \cdot \text{ SE of } \hat{\theta}$$

Example: Odds ratio. From $\hat{p}$, compute $\hat{\text{OR}} = \hat{p}/(1-\hat{p})$.

- $g(p) = p/(1-p), \quad g'(p) = 1/(1-p)^2$
- $\text{SE}(\hat{\text{OR}}) \approx \frac{1}{(1-\hat{p})^2} \cdot \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} = \frac{\sqrt{\hat{p}}}{(1-\hat{p})^{3/2}\sqrt{n}}$

The Delta Method: CIs for Transformations

Problem: You have a CI for θ, but you want a CI for $g(\theta)$.

Delta method: If $\hat{\theta} \dot{\sim} N(\theta, \sigma^2_{\hat{\theta}})$, then for smooth g:

$$g(\hat{\theta}) \quad \dot{\sim} \quad N\big(g(\theta), \quad [g'(\theta)]^2 \cdot \sigma^2_{\hat{\theta}}\big)$$

$$\text{SE of } g(\hat{\theta}) \quad \approx \quad |g'(\hat{\theta})| \cdot \text{ SE of } \hat{\theta}$$

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Tip: Often easier to build CI on the log scale, then exponentiate the endpoints. This gives asymmetric CIs, which are usually more appropriate for ratios.

Summary: From Point to Interval

Sampling distribution: $\hat{\theta}$ varies across samples. Its distribution governs uncertainty.

Standard error: $\mathrm{SE} = \mathrm{SD}(\hat{\theta})$. For the mean: $\sigma/\sqrt{n}$. For any MLE: $1/\sqrt{nI(\hat{\theta})}$.

The $\sqrt{n}$ law: Halving the SE requires quadrupling n. Precision is expensive.

Wald CI: $\hat{\theta} \pm z_{\alpha/2} \cdot \mathrm{SE}$. Works when $\hat{\theta}$ is approximately Normal.

Interpretation: 95% of *intervals* contain θ, not "95% chance θ is here."

Proportions: Use Wilson, not Wald, when $n\hat{p} < 5$ or $\hat{p}$ is near 0/1.

CI vs credible interval: Frequentist vs Bayesian. Same answer for large n.

Delta method: $\mathrm{SE}(g(\hat{\theta})) \approx |g'(\hat{\theta})| \cdot \mathrm{SE}(\hat{\theta})$. CIs for transformations.

Practical: Building Confidence Intervals

1. **CI coverage simulation:**
   - Generate $n = 30$ samples from $N(\mu, \sigma^2)$, compute 95% CI
   - Repeat 10,000 times. Count how many CIs contain μ
   - Verify: should be $\approx$ 95%. Try $n = 5$ — still 95%?
2. **Wald vs Wilson for proportions:**
   - Simulate $n = 20$, $p = 0.05$. Compare coverage of Wald and Wilson
   - Plot coverage as a function of p for both methods
3. **CI for Poisson:**
   - Compute 95% CI for λ using Fisher information
   - Compare with bootstrap CI (preview of Lecture 8)

Homework

1. A factory tests $n = 100$ lightbulbs: $\bar{X} = 1{,}150$ hours, $S = 200$ hours.
   Construct 90%, 95%, and 99% CIs for the true mean. Which is widest? Why?

2. A poll surveys $n = 500$ voters: 265 support candidate A ($\hat{p} = 0.53$).
   (a) Compute the Wald 95% CI. Can we declare A is leading?
   (b) How many voters do we need for a $\pm 2\%$ margin of error?

3. For $X_1, \ldots, X_n \sim \text{Exp}(\lambda)$, the MLE is $\hat{\lambda} = 1/\bar{X}$.
   (a) Derive the SE using Fisher information ($I(\lambda) = 1/\lambda^2$).
   (b) Use the delta method to find a CI for the mean $\mu = 1/\lambda$.
   (c) Compare with a direct CI for μ using $\text{SE}(\bar{X}) = \bar{X}/\sqrt{n}$.

4. Simulate 1,000 Wald 95% CIs for a proportion with $n = 15$, $p = 0.02$.
   What fraction contain p? Repeat with Wilson. Which is closer to 95%?

Recommended Visualizations & Resources

Interactive: Confidence Interval Simulation (R Psychologist)

`rpsychologist.com/d3/ci` — watch CIs accumulate in real time. Drag sliders for n, σ, confidence level. The best CI demo.

Interactive: Seeing Theory — Frequentist Inference (Brown)

`seeing-theory.brown.edu/frequentist-inference` — point estimation, CI, and hypothesis testing with beautiful animations.

Video: StatQuest — Confidence Intervals

`statquest.org/confidence-intervals` — clear walkthrough of what CIs are and are not. Great for correcting misconceptions.

Interactive: Sample Size Calculator

`sample-size.net` — plan studies interactively. Set margin of error, confidence level, and see required n.

Reading: Penn State STAT 415 — Confidence Intervals

`online.stat.psu.edu/stat415/lesson/2` — worked examples for Normal, Binomial, and Poisson CIs with derivations.

Questions?

Next: Lecture 8 — Bootstrap and resampling methods